

Unfolded Protein Binding Annotation Review

Reclassifying GO:0051082 & GO:0031249

Chris Mungall | AI-Assisted Gene Review
2026-10-04 refresh

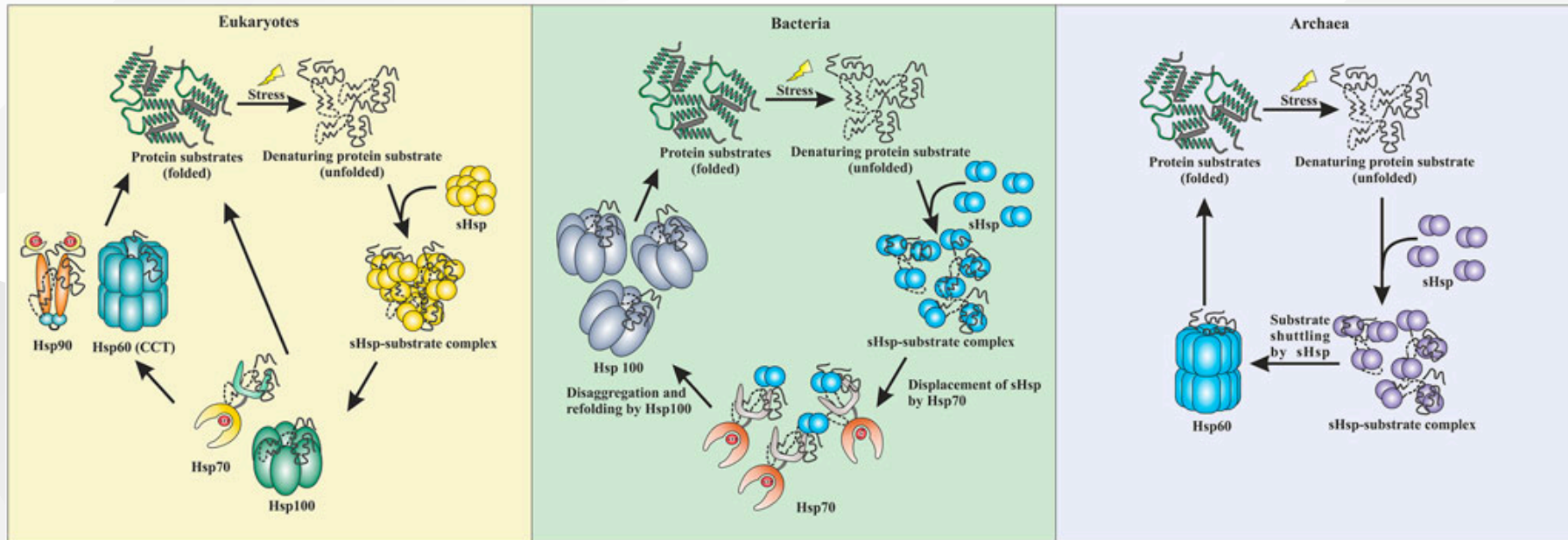
The Problem

GO:0051082 "unfolded protein binding" and **GO:0031249** "denatured protein binding" were obsoleted ([go-ontology#30962](https://www.ebi.ac.uk/GO/help/faq/faq30962.html))

These terms conflate mechanistically distinct activities under a single "binding" label:

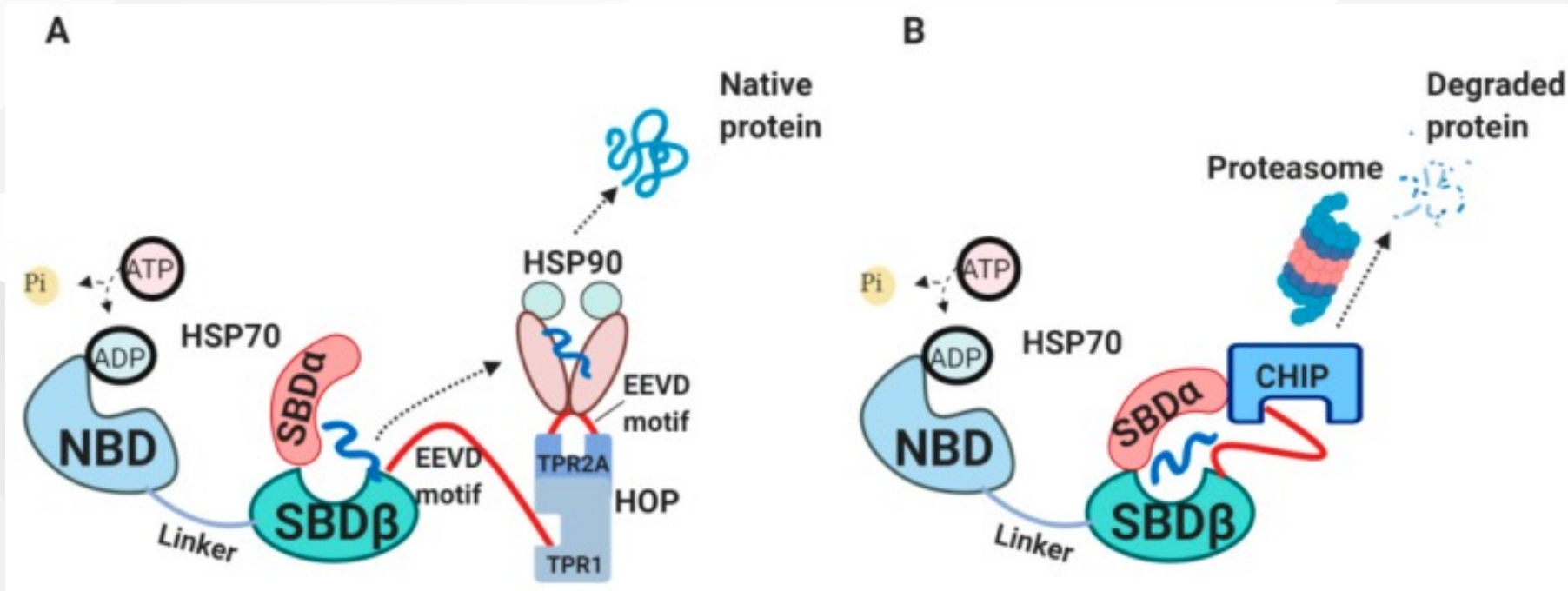
- ATP-dependent **foldases** that actively refold clients
- ATP-independent **holdases** that prevent aggregation in situ
- **Disaggregases** that solubilize existing aggregates
- **Sensors** that recognize misfolded proteins for degradation
- **Co-chaperones** that modulate chaperone ATPase cycles

The Holdase-Foldase Network



Conserved across all domains of life: stress unfolds proteins; **sHSPs (holdases)** capture them to prevent aggregation; clients are handed off to **Hsp70 (foldase)** for ATP-dependent refolding, or to **Hsp100 (disaggregase)**. Eukaryotes also use Hsp90 and Hsp60/CCT.

The Proteostasis Triage Decision



(A) HSP70-HOP-HSP90 axis: fold-forward pathway to native protein.

(B) HSP70-CHIP axis: ubiquitination and proteasomal degradation.

Co-chaperones (DNAJ, NEFs, BAG1/3) tune the **fold vs degrade** decision.

Recent Literature Context

Key reviews informing this reclassification:

- **Mitra et al. (2022)** *Annu Rev Biophys* 51:409 — ATP-independent chaperones; force-clamp experiments show context-dependent holdase/foldase switching
- **Bhattacharjee et al. (2025)** *J Biosci* — Co-chaperones fine-tune HSP70: fold, hold, or degrade; O-GlcNAc modification of HSP27 promotes BAG3-mediated refolding
- **Boopathy (2024)** — HUWE1 + HSP70 cooperate to clear nuclear inclusions, linking holdase activity to E3-mediated clearance
- **Le et al. (2024)** *Mol Cells* — UBR1/UBR2 as ER-stress sensors: substrate load stabilizes these E3s to enhance PQC

Phase-1 Scope

Metric	Count
Unique genes in frozen audit	148
Human genes (primary)	33
Non-human genes	115
Species covered	17
Total annotations reviewed	5,529
Remaining PENDING	0

Feb 2026 action snapshot; the live registry has since grown through follow-up case studies and identity corrections.

Mechanism Classes

Class	GO term	ATP?	Example
Foldase	GO:0044183	Yes	GroEL/ES, TRiC/CCT
Holdase	<i>NTR needed</i>	No	CRYAB, CLU
Carrier-holdase	GO:0140309	No	Tim9-Tim10
Foldase/holdase	GO:0044183 + holdase NTR	Yes	HSPA1A (HSP70)
Co-chaperone	<i>(see gap)</i>	N/A	DNAJB1, AHSA1
Disaggregase	GO:0140545	Yes	HSP104, HSPA1A
Sensor	<i>NTR proposed</i>	N/A	SYVN1, UGGT1

Decision Rules

Mechanism	Action	Replacement
Foldase (GroEL, TRiC)	MODIFY	GO:0044183
Foldase/holdase (HSP70)	MODIFY	GO:0044183 (<i>holdase NTR pending</i>)
Co-chaperone, J-domain	MODIFY	GO:0044183 (<i>interim</i>)
Holdase (sHSP, crystallin)	MODIFY	proposed holdase chaperone activity
Disaggregase	MODIFY	GO:0140545
ER/QC sensor	REMOVE	(<i>not chaperones</i>)
HSP90 co-chaperone	OVER ANNOTATED	(<i>not direct UPB</i>)

Impact: Human Genes (n=33)

Primary action	Count	Notes
MODIFY to GO:0044183 (foldase)	16	HSP70 (6), J-domain interim (4), prefoldin (6)
MODIFY to proposed holdase chaperone activity	6	sHSPs, CLU, DNAJB6, DNAJB8
MODIFY to other specific MF	3	NPM1, AIP, SCG5
MARK_AS_OVER_ANNOTATED	5	Sensor/co-chaperone cases
REMOVE	3	SYVN1, ERLEC1, GRPEL1

Plus 3 co-annotations to [GO:0140545](#) (disaggregase): HSPA1A, HSPA1B, HSPA8

Before/After Examples

Gene	Before	After	Why
HSPA1A	GO:0051082	GO:0044183 + GO:0140545	ATP-dependent foldase + disaggregase
CRYAA	GO:0051082	proposed holdase chaperone activity	sHSP holdase; prevents aggregation in situ
DNAJB1	GO:0051082	GO:0044183 (<i>interim</i>)	J-domain co-chaperone, not independent foldase
SYVN1	GO:0051082	REMOVE	E3 ligase; recognizes misfolded substrates
NPM1	GO:0051082	GO:0140713	Histone chaperone; UPB was secondary

Critical Finding: The Holdase Gap

GO:0140309 "unfolded protein holdase activity" does **not** fit most holdases.

- Created Nov 2025 for TIM carrier-holdases (Tim9-Tim10) in [go-ontology#30552](#)
- Definition still requires escort "to an acceptor molecule or to a specific location"
- It remains a child of GO:0140597 "protein carrier chaperone"
- The primary **holdase** label outgrew the carrier-specific definition

6 human genes + HSPH1 are in-situ holdases that prevent aggregation without delivery to a defined acceptor or destination. They cannot use GO:0140309.

Proposed: "Holdase Chaperone Activity"

Definition: Binding to an unfolded or misfolded protein to prevent its aggregation without actively catalyzing refolding. The holdase maintains the client protein in a soluble, folding-competent state.

Parentage: Direct child of GO:0003674 (molecular_function).

[GO:0140309](#) (carrier-holdase) becomes a child of this new term.

Affected genes: CRYAA, CRYAB, HSPB6, CLU, DNAJB6, DNAJB8, HSPH1

Remaining in-situ holdase migrations are tracked in [ai-gene-review#2222](#).

Other Ontology Gaps

Misfolded protein sensor activity

- Recognition of misfolded conformation for QC degradation
- CHIP/STUB1 ubiquitinates chaperone-bound clients; UBR1/UBR2 recognize N-degrons ([Le et al. 2024](#))
- Affects: SYVN1, SAN1 (yeast), Fbxo2 (mouse)

Co-chaperone MF representation

- [GO:0003767](#) "co-chaperone activity" deliberately obsoleted
- No MF term for J-domain co-chaperone function (ATPase activation + substrate delivery)
- [GO:0044183](#) used as pragmatic interim
- Affects all J-domain proteins across species

Phase-1 Cross-Species Stress Test

The phase-1 audit applied the human-derived categories across 17 species and exposed bacterial, fungal and metazoan edge cases.

Species	Genes	Highlights
<i>S. cerevisiae</i>	67	Most mechanism classes represented
<i>E. coli</i>	13	Periplasmic holdases (SurA, Skp, Spy, HdeA/B)
<i>D. melanogaster</i>	7	sHSPs Hsp22-27
<i>M. musculus</i>	6	Hspa8 largest review (240 annotations)
<i>R. norvegicus</i>	4	Hspa5/BiP (101 annotations)
Other (12 spp.)	18	Zebrafish crystallins, CnoX redox holdase

Notable Cross-Species Findings

- **SlrP** (*S. typhimurium*): **misannotation removed** — T3SS effector E3 ligase, not a chaperone
- **CnoX** (*E. coli*): redox-activated holdase — becomes active under oxidative stress when Cys residues are oxidized
- **Peroxiredoxins** (TSA1, pmp20, tpx1): dual-function enzymes with stress-dependent holdase activity
- **Assembly factors** (ATP10, PET100, SHY1): single-client chaperones, not general UPB
- **IRE1** (yeast + *T. reesei*): UPR sensor, not chaperone — cross-kingdom confirmation

What We Need from GO Editors

1. **Holdase NTR (UNRESOLVED)**: Create "holdase chaperone activity" for in-situ holdases
2. **Repair obsolescence guidance** for obsolete GO:0051082 / GO:0031249
3. **Preferred labels**: "foldase" exact synonym on GO:0044183; reserve plain "holdase" for the general term
4. **Co-chaperone MF gap**: How should J-domain function be annotated?
5. **HSP70 dual annotation**: Confirm foldase + holdase per experimental context
6. **Misfolded protein sensor**: New term for SYVN1, SAN1, Fbxo2?

Summary

148 phase-1 genes across **17 species**, **5,529 annotations** reviewed

The key bottleneck is the **holdase NTR** — without it, six human in-situ holdases plus HSPH1 lack a valid GO replacement after GO:0051082 and GO:0031249 obsolescence.

All gene review YAMLS: `genes/<SPECIES>/<GENE>/`

Validate: `just validate-all`